Reg. No															

B.Tech/M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

Sixth Semester

21CSS303T - DATA SCIENCE

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

		Marks	BL	CO
1.	The ______ is the primary goal of Data Science? (A) To store large amounts of data (B) To analyze and extract insights from data (C) To design websites (D) To create databases	1	2	1
2.	Which of the following operations is faster in NumPy than in Python lists? (A) Looping through elements one by one (B) String manipulations (C) Defining functions (D) Vectorized operations	1	2	1
3.	The _______ function is used to create a NumPy array? (A) array() (B) create_array() (C) list_to_numpy() (D) make_array()	1	1	2
4.	What is the primary difference between a Pandas Series and DataFrame? (A) Series holds multiple columns, DataFrame holds a single column (B) Series is one-dimensional, DataFrame is two-dimensional (C) DataFrame is immutable, Series is mutable (D) Series does not support indexing	1	2	2
5.	________ function is used to remove missing values in Pandas? (A) dropna() (B) fillna() (C) replace() (D) remove_nan()	1	2	3
6.	The main challenge in handling large datasets _______. (A) Storing the dataset in Excel (B) High computational cost and memory usage (C) Creating charts (D) Adding more rows manually	1	1	3
7.	Which Pandas function is used to merge two DataFrames on a common column? (A) merge() (B) concat() (C) join() (D) combine()	1	1	3
8.	To detect outliers in a dataset the ________ is used. (A) Min-Max Scaling (B) Standardization (C) Interquartile Range (IQR) (D) Label Encoding	1	2	3
9.	The purpose of Web Scraping is ________. (A) To manually download data from websites (B) To improve website SEO rankings (C) To delete unwanted data from a database (D) To extract structured data from web pages automatically	1	2	3

		Marks	BL	CO
10.	Which of the following functions is used to change the index of a DataFrame? (A) reset_index() (B) reindex() (C) set_index()reindex() (D) reset_index() and set_index()	1	1	4
11.	The _______ library is most commonly used for data visualization in Python? (A) NumPy (B) Matplotlib (C) Pandas (D) Scikit-learn	1	1	4
12.	What type of plot is best for showing the distribution of a numerical variable? (A) Bar Plot (B) Scatter Plot (C) Histogram (D) Pie Chart	1	1	4
13.	The purpose of a Box Plot is _______. (A) To display the spread of a dataset and detect outliers (B) To show trends over time (C) To visualize categorical variables (D) To compare two different datasets	1	2	4
14.	The ______ function is used to read a CSV file in Pandas? (A) pd.open_csv() (B) pd.read_csv() (C) pd.load_csv() (D) pd.get_csv()	1	2	3
15.	What type of chart is best for showing trends over time? (A) Bar Chart (B) Pie Chart (C) Line Chart (D) Scatter Plot	1	2	4
16.	The ______ Python library is commonly used for web scraping? (A) NumPy (B) Pandas (C) BeautifulSoup (D) Scikit-learn	1	1	2
17.	To sort a Pandas DataFrame by a specific column, the ________ function is used. (A) df.sort() (B) df.orderby() (C) df.arrange() (D) df.sort_values()	1	2	3
18.	Identify the function used to check the first few rows of a DataFrame? (A) df.top() (B) df.head() (C) df.first() (D) df.show()	1	2	3
19.	Which function is used to remove duplicate rows in a Pandas DataFrame? (A) drop_duplicates() (B) remove_duplicates() (C) clear_duplicates() (D) delete_duplicates()	1	1	3
20.	The _______ function is used to create an array of zeros in NumPy. (A) np.empty() (B) np.ones() (C) np.zeros() (D) np.full()	1	1	4

PART - B (4 × 10 = 40 Marks)

Answer **any 4** Questions

		Marks	BL	CO
21.	The following NumPy array representing daily temperature readings: import numpy as np temps = np.array([30, 32, 28, 35, 31, 29, 34]) (a) Write Python code to calculate the mean, maximum, and minimum temperature using NumPy. *(5 Marks)* (b) Write Python code to sort the temperatures in ascending order and find the index of the highest temperature. *(5 Marks)*	10	3	3
22.	(a) What is Web Scraping? Explain its applications and ethical concerns. *(5 Marks)* (b) Compare Web APIs and Open Data Sources as methods of data acquisition in Data Science. *(5 Marks)*	10	4	2

23. The following dataset representing house prices | 10 | 3 | 3

```
import pandas as pd
data = {'Size (sq ft)': [1000, 1500, 2000, 2500, 3000],
   'Price ($)': [150000, 200000, 250000, 300000, 350000]}
df = pd.DataFrame(data)
```

(a) Write Python code to create a scatter plot to visualize the relationship between house size and price. *(5 Marks)*
(b) Use Seaborn to fit a regression line to the scatter plot. *(5 Marks)*

24. (a) Describe a step-by-step approach to identify and handle duplicate records and missing values. *(5 Marks)* | 10 | 2 | 3
(b) Explain the importance of data standardization and discuss two techniques used to normalize data. *(5 Marks)*

25. (a) What are the different types of data visualizations used in Data Science? Explain any three with examples. *(5 Marks)* | 10 | 4 | 4
(b) Discuss how Matplotlib and Seaborn can be used for customizing plots. Mention three key customization techniques. *(5 Marks)*

26. Using Seaborn, write Python code to: | 10 | 3 | 4
(a) Create a scatter plot between "Marketing Spend" and "Sales Revenue" using a dataset. *(5 Marks)*
(b) Use a regression line to show the trend between the two variables. *(5 Marks)*

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Questions

Marks BL CO

27. **a)** Discuss the challenges faced when handling large datasets in data analysis. Provide three general techniques to efficiently manage and process large volumes of data. (5 marks) | 15 | 4 | 3
b) Explain the following data wrangling operations with examples: (5 marks)
- Merging datasets on an index
- Concatenating datasets
- Reshaping data using pivoting

c) You are given a dataset with the following columns: Name, Age, Salary, and Department. The dataset contains missing values in the Salary column and some outliers in the Age column. Write a Python code snippet using Pandas to: (5 marks)
- Fill missing Salary values with the median salary of the respective department.
- Identify and remove outliers in the Age column using the IQR (Interquartile Range) method

28. You are a data scientist working on a project that analyzes monthly sales data for an e-commerce company. The dataset includes total sales revenue, the number of customers, product categories, and discount percentages over the last three years. The company wants insights into: | 15 | 4 | 5
   1. Seasonal trends in sales using line and bar plots.
   2. The impact of discounts on sales revenue using scatter plots.
   3. Customer purchasing patterns using histograms and box plots.
   4. Correlations between product categories and revenue using pair plots.
   5. Improving presentation quality by customizing plots (titles, labels, colors, and styles).

a) Explain the role of different types of plots in understanding e-commerce sales data. Specifically, describe how a line plot, scatter plot, and histogram can provide insights into sales trends. *(5 Marks)*

(b) You are required to create a figure with multiple subplots to display monthly sales trends for different product categories. Explain how you would use Matplotlib to create four subplots in a single figure, including details on layout, labels, legends, and customization. *(5 Marks)*

(c) To improve the presentation quality of your visualizations, discuss three techniques for customizing plots using Matplotlib and Seaborn. *(5 Marks)*
   1. Using different plot styles and themes.
   2. Modifying font sizes, tick labels, and legends for better readability.
   3. Adding annotations and markers to highlight key trends.

*****